



KITTISAK TANTRATONE



SCAN MY WEBSITE

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PROFESSIONAL SUMMARY

CS student specializing in GIS & Remote Sensing, AI Pipelines, and IoT systems. Experienced in developing Machine Learning and Neural Network models for multi-modal satellite data analysis and agricultural decision support. Skilled in building automated multimedia systems using Generative AI and NLP, with a foundational background in Cybersecurity from national CTF competitions and Hackathons.

EDUCATION

KING MONGKUT'S UNIVERSITY OF TECHNOLOGY NORTH
BANGKOK (KMUTNB)
BACHELOR OF SCIENCE IN COMPUTER SCIENCE | GPAX: 2.93

SELECTED PROJECTS

GISXKMUTNB: MULTI-MODAL SATELLITE PALM DETECTION (SENIOR PROJECT, IN-PROGRESS)

- **MECHANISM:** DEVELOPING A SELF-SUPERVISED (MAE) + FINE-TUNING PIPELINE ON A VISION TRANSFORMER (VIT) ARCHITECTURE TO DETECT OIL PALM PLANTATIONS. FUSES MULTI-SENSOR SATELLITE DATA (SENTINEL-1 SAR, SENTINEL-2 OPTICAL, AND GEO-SATELLITE RGB) INTO A 12-BAND INPUT TENSOR, OUTPUTTING PIXEL-WISE PROBABILITY MAPS. BUILT AS TWO REUSABLE PYTHON LIBRARIES (DATA FUSION + VIT MODEL) WITH A PROOF OF CONCEPT TRAINING PIPELINE ON WEBSITE.

AI-ASSISTED PALM DECISION SUPPORT SYSTEM (DSS) (IN-PROGRESS)

- **MECHANISM:** CONTRIBUTING TO THE AI PIPELINE FOR A PALM DECISION SUPPORT SYSTEM. APPLIED SAM (SEGMENT ANYTHING MODEL) TO SEGMENT PLANTATION AREAS, AND USED GEOAIPY TO CLASSIFY PALM COVERAGE AND GENERATE PROBABILITY MAPS TO SUPPORT DECISION-MAKING.

AI NOVEL NARRATOR & ILLUSTRATIVE BACKGROUND (ON HOLD – PERSONAL HOBBY PROJECT)

- **MECHANISM:** AUTOMATED MULTIMEDIA PIPELINE USING DEEPSEEK LOCAL FOR TEXT STRUCTURING AND CHARACTER DIALOGUE IDENTIFICATION, WANGCHANBERTA FOR EMOTION DETECTION FEEDING INTO A VOICE CLONING SYSTEM (OPENVOICE), AND ANIMAGINE XL 4.0 FOR CONTEXT-AWARE IMAGE GENERATION. CURRENTLY PAUSED TO FOCUS ON PRIMARY GIS RESEARCH WORK.

SELECTED PROJECTS

AI-BASED MOTORCYCLE RIDER HELMET DETECTION

- **MECHANISM:** APPLIED OBJECT DETECTION (YOLO/TENSORFLOW) FOR REAL-TIME VIDEO PROCESSING TO VERIFY AND LOCATE HELMET USAGE FOR ROAD SAFETY.

POLYDRY PREDICTOR (IOT FOR 3D PRINTING)

- **MECHANISM:** INTEGRATED DHT22 SENSORS WITH AN MCU TO TRANSMIT REAL-TIME HUMIDITY AND TEMPERATURE DATA TO A SERVER FOR CALCULATING OPTIMAL FILAMENT DEHYDRATION.

MOBILE PRICE PREDICTION SYSTEMS

- **NEW MOBILE (2020-2025):** IMPLEMENTED A REGRESSION MODEL ANALYZING HARDWARE SPECIFICATIONS AND HISTORICAL DATA TO PREDICT CURRENT MARKET PRICE TRENDS.
- **OLD MOBILE (2000-2010):** ENGINEERED A NEURAL NETWORK TO PROCESS COMPLEX, NON-LINEAR DATA RELATIONSHIPS FOR CLASSIC MOBILE PHONE VALUATION.

TECHNICAL SKILLS

- **GIS & Remote Sensing:** QGIS, Python (Geopandas, Rasterio), SNAP (Sentinel Application Platform), GeoAI / GeoAlpy, Multi-Modal Satellite Data Fusion (Sentinel-1 SAR, Sentinel-2 Optical), Segment Anything Model (SAM)
- **Artificial Intelligence:** Machine Learning (Regression), Neural Networks (MLP), Computer Vision (Object Detection), Generative AI (TTI, LoRA, TTS)
- **IoT & Hardware:** Microcontrollers (MCU), 3D Printer Optimization (Bambu Lab A1 ecosystem), Raspberry Pi
- **Programming:** Python, C++, Java, JavaScript, SQL, HTML, CSS, Shell Scripting
- **Cybersecurity (Foundational):** Penetration Testing basics, Digital Forensics; familiar with Kali Linux, Nmap, Wireshark, Burp Suite

ACHIEVEMENTS & ACTIVITIES

- **CYBERSECURITY COMPETITIONS:** ACTIVE PARTICIPANT IN CAPTURE THE FLAG (CTF) COMPETITIONS, ACHIEVING A TOP 380 INDIVIDUAL RANKING. 211 TEAM RANKING, EXPERIENCED IN CYBER HACKATHONS FOCUSING ON DIGITAL FORENSICS AND VULNERABILITY ASSESSMENT.
- **MAKER & DIY PROJECTS:** DISCOVERING AND CREATING FUNCTIONAL 3D MODELS SPECIFICALLY ENGINEERED TO SOLVE MINOR DAILY. EXPERT IN HARDWARE TROUBLESHOOTING AND MODIFICATION FOR BAMBU LAB A1 PRINTERS.